

JAKE LEEUWENBURG

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EDUCATION

California Polytechnic State University, San Luis Obispo
Bachelor of Science in Mechanical Engineering

Sep. 2023
San Luis Obispo, CA

EXPERIENCE

Research and Development Engineer
Innovative Drive Corporation

Jul. 2025 – Present
Reno, NV

- Developed the mechanical design of a novel intravascular lithotripsy (IVL) device from concept through verification
- Advanced the device to successful first-in-human cases and delivered verification data for the FDA IDE submission
- Designed injection-molded and machined components; drove DFM iterations with an international injection molder
- Built benchtop test models of clinical and worst-case use conditions, generating data that drove design decisions
- Authored DHF/DMR documentation, drawings, protocols, and reports per ISO 13485 / 21 CFR 820 design controls
- Executed design reviews, design V&V, DFMEA, and PFMEA under ISO 14971 risk management processes
- Supported animal and cadaver studies, sterilization, IEC 60601 safety/EMC, and packaging transit conditioning testing
- Integrated Hall effect sensors and magnets, IR optical sensors, and a fluid pressure sensor into the device design
- Instrumented bench tests with dynamic (explosion) and low-frequency pressure sensors and oscilloscope data capture
- Prepared design transfer to a contract manufacturer via DFM/DFA, component specifications, and assembly fixtures
- Resolved key issues on tight customer deadlines with Electrical, Software, Quality, Clinical, and Regulatory teams

Quality and Manufacturing Engineer
Inspire Products Inc.

Jan. 2024 – Jul. 2025
Reno, NV

- Designed nitinol shape-setting fixtures that cut heat-cycle time 56%, changeover time 51%, and raised yield 29%
- Authored and executed OQ/PQ protocols to qualify manufacturing processes for medical device components
- Built and validated electro-mechanical inspection fixtures and trained technicians, improving outgoing quality
- Led CAPA investigations and root cause analyses; implemented corrective actions and authored reports
- Maintained the ISO 13485 / ISO 9001 QMS, supported internal and supplier audits, and ran SPC and Cp/Cpk studies

PROJECTS

Cam-Driven Hammer and Die Spring Simulator | JavaScript, Three.js, Blender

- Built an interactive 3D simulation to diagnose spring rotation and seat scoring in a cam-driven hammer mechanism
- Identified root cause and vetted the design change in simulation, with only a few test units available to verify the fix

IVL Fluid-Path Acoustic Simulator | JavaScript, 1-D finite-difference solver

- Wrote a 1-D acoustic wave solver (Webster's horn equation) to predict pressure pulse transmission in the fluid path
- Evaluated fluid-path geometry, wall compliance, and cavitation risk to guide design decisions ahead of bench testing

Design Verification Data Analysis Toolkit | Python, pandas, SciPy, matplotlib

- Built GUI and batch tools that score leak, pressure range, and hit frequency tests and generate an HTML dashboard
- Developed a transient spectral analysis pipeline (STFT, peak detection, calibration) to analyze device performance

TECHNICAL SKILLS

Design & Manufacturing: SolidWorks, GD&T, DFM/DFA, injection molding, machining, fixture design, 3D printing

Quality & Regulatory: Design controls, ISO 13485/9001, ISO 14971, DFMEA/PFMEA, V&V, OQ/PQ, CAPA, SPC/Cpk

Testing & Instrumentation: Benchtop test models, pressure transducers, Hall effect and IR sensors, oscilloscopes, OMM

Materials & Processes: Nitinol shape setting, heat staking, soldering, plastic and metal component specification

Programming & Tools: Python (pandas, SciPy), JavaScript, Arduino, AI-assisted development (Claude Code)

Certification: ASQ Certified Quality Improvement Associate (CQIA)